

Inventory Management System

Development Intern Task — 2 Days

Overview

Build a simple Inventory Management System that helps track products, manage stock levels, and view inventory statistics. This is a frontend-only project — no backend or database is needed. Use `localStorage` to store and persist all data.

Tech Stack

- React.js (not mandatory)
- Formik + Yup — for form handling and validation (mandatory)
- Any UI/styling library of your choice (MUI, Ant Design, Tailwind CSS, etc.)
- `localStorage` — for data persistence (no backend required)

Features to Implement

1. Product Management

Users must be able to fully manage products in the inventory.

- Add a new product
- Edit an existing product
- Delete a product
- Each product must have the following fields:
 - Product Name
 - Product ID
 - Category
 - Price
 - Stock Quantity

2. Stock Management

Stock levels must be updated accurately and safely.

- Increase stock quantity (restock / incoming)
- Decrease stock quantity (sales / outgoing)

3. Dashboard / Stats View

A summary screen giving a quick overview of the inventory.

- Ex: Total number of products and Total inventory value

4. Category Handling

Products must be organized and manageable by category.

- Assign a category to each product
- Allow users to create custom categories
- Filter products by category
- Show product count per category on the dashboard

5. Search & Filter

Users should be able to quickly find products.

- Search by product name or SKU
- Filter by category
- Filter by stock status (In Stock / Out of Stock)

6. UI Requirements

- Clean and simple layout — easy to navigate
- Responsive — works on both mobile and desktop
- Table or card view for the product list
- Use Formik + Yup for all forms with proper error messages

Bonus Features (Optional)

These are not required but will improve your evaluation score.

Feature	Description
Auto-generated SKU	Generate a unique Product ID automatically (e.g. PRD-482910)
Stock History Log	Record every stock change with a timestamp
Export to CSV	Download the full product list as a .csv file
Dark Mode	Add a dark/light theme toggle, saved in localStorage
Analytics Chart	Show a chart of stock or category distribution
Bulk Actions	Select multiple products to delete or restock at once

Evaluation Criteria

1. Code Quality — clean, readable, and well-structured code
2. Functionality — all core features work correctly, edge cases handled
3. User Experience — responsive, easy to use, clear feedback
4. Form Validation — Formik + Yup used correctly with useful error messages
5. Git Commit History — frequent, meaningful commits that show your progress
6. Bonus Features — extra credit for well-implemented optional features

For Git commits, use clear messages that describe what was done, for example:

- feat: add product form with Formik validation
- fix: prevent stock from going below zero
- feat: implement category filter
- feat: add CSV export

Submission Guidelines

7. Submit a link to a public GitHub repository
8. Include a README.md with:
 - Project overview
 - How to run the project locally
 - List of features implemented
 - Screenshots of the application
9. (Optional) Live deployment link — Vercel or Netlify recommended

Important Note

Candidates are expected to understand and be able to explain their implementation. Submissions must reflect your own original work.